

- The order
- Product development team: Throwing ideas
- ~~Tier 1~~ Airplane group, has subsystems such as safety, flight control, flight deck, landing gear, stress, aerodynamic & these groups create Tier 1. Then goes go down to avionics, ^{Tier 2} other groups breaking from that main topic & then to Tier 3 such as display, & other components, called SCD (Specification) then goes to supplier for them to do requirements on the actual part

LRU gear data from antenna ^{process} from analog to digital & send to display

ATC Aircraft Traffic Controller

Nano 10^{-9}
 micro 10^{-6}
 milli 10^{-3}
 centi 10^{-2}
 kilo 10^3
 mega 10^6
 giga 10^9

Notes from DBP Chb

$$1 \frac{\text{Available power} - P_{\text{required to overcome aero. drag}}}{\text{weight}}$$

$$C_D = C_{D0} + \frac{C_L^2}{\pi A e} \text{ Parasite Drag}$$

Eq-n of motion

$$-\Sigma F_{||} = T \cos \theta - W \sin \theta - D = ma \text{ (x axis)}$$

$$-\Sigma F_{\perp} = T \sin \theta + L - W \cos \theta = m \frac{v^2}{r_e} ?$$

$$1 \text{ Thrust requirement} = \frac{W}{C_L / C_D} = \frac{W}{L/D} \text{ for } \text{st} \text{ level flight}$$

Given V_{∞} , C_L for stabilization can be found by

$$- C_L = \frac{W}{0.5 \rho_{\infty} S V^2}$$

Do the same for C_D to create ratio $\frac{C_L}{C_D}$ to calc. thrust requirements

Power enough to stabilize, R-requirements

$$P = \frac{\text{energy}}{\text{time}}, P_R = T_{\text{requirements}} \cdot V_{\infty}$$

Ribs ~~can~~ amount can be determined by simulation of the by FEA software

$$1 q_{\text{bar}}$$

$$- q = \frac{1}{2} \rho S V^2$$

$$V_{\text{takeoff}} = \sqrt{\frac{2W}{\rho S C_L}}$$